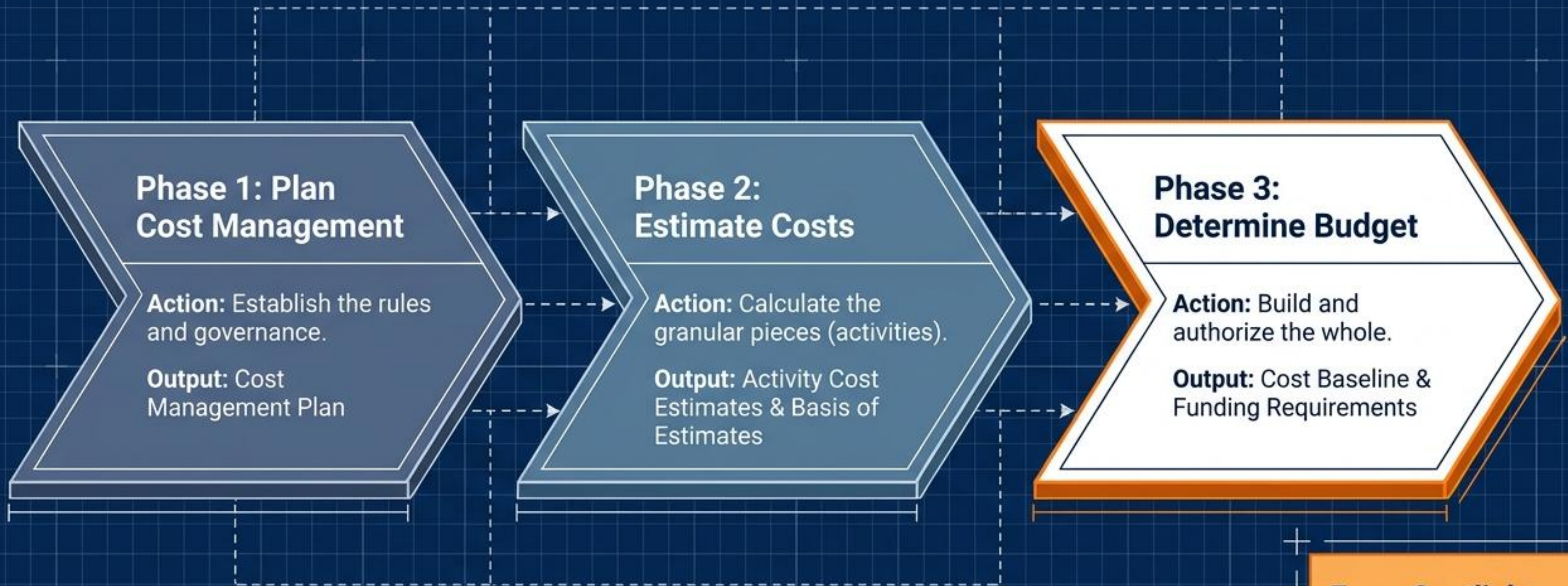


Certyfikat PMP: Project Cost Management



Constructing the Authorized Budget:
A Master Schematic for PMBOK® Methodology

dr Zbigniew Galar



Exam Spotlight

Master the Flow: Activity estimates from Phase 2 are the direct input required to build the baseline in Phase 3.

Phase 1: Drafting the Financial Rules (Plan Cost Management)

Accuracy & Measurement



Level of Accuracy
(e.g., $\pm 5\%$, rounding)



Units of Measure
(hours, days, lump sum)

Organizational Links



Control Thresholds

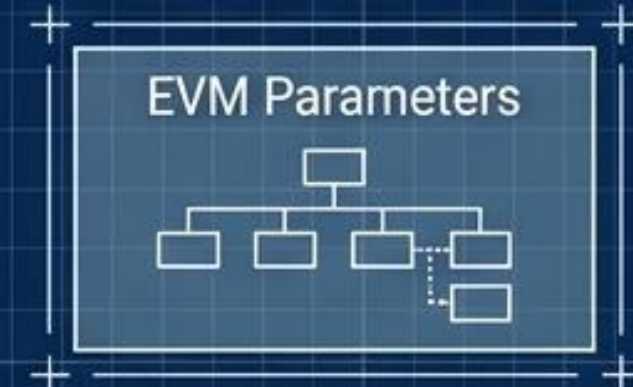


Allowable % deviation before
sponsor action is triggered.

Exam Spotlight:

The Cost Management Plan doesn't contain actual costs—it dictates HOW you will measure and control them.

Performance Rules



Defines where and how Earned Value Management will be measured in the WBS.

Phase 2: Calculating the Pieces (Estimate Costs)

Work Package

Resource Costs

Human labor, materials, equipment, facilities, inflation factors.

Trade-off Analysis

Make-vs-Buy, Buy-vs-Lease, Resource Sharing evaluations.

Cost of Quality (COQ)

The price of conforming to established project standards.

Life Cycle Costs

Recurring maintenance and ongoing support post-project.

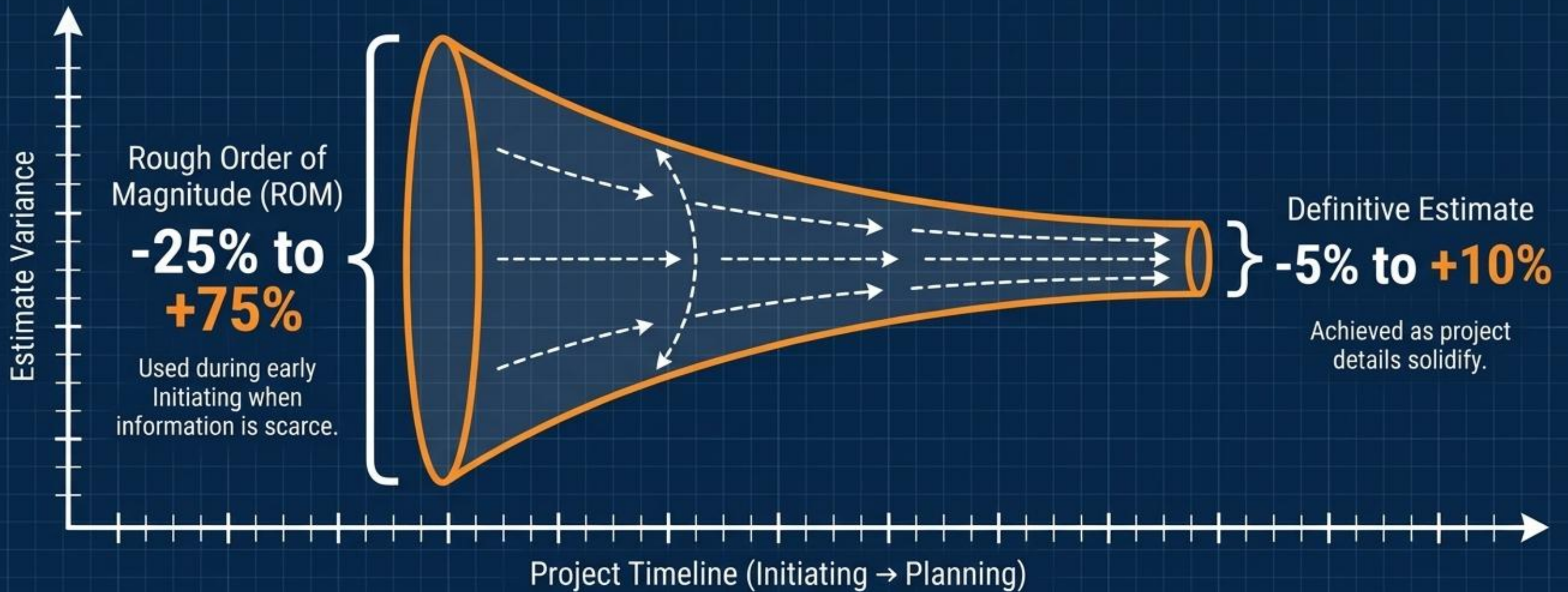
Exam Spotlight:

Pricing ≠ Costing. Estimate Costs focuses solely on the resources needed to produce the project, excluding your company's profit margin.

The Estimation Toolkit Matrix

	Analogous (Top-Down)	Parametric	Bottom-Up
Concept	Uses historical actuals from similar projects via Expert Judgment.	Quantitative relationship (e.g., hours × resource rate).	Estimates every individual activity and rolls them up.
Project Phase	Early / Initiating.	Planning.	Detailed Planning.
Accuracy Level	Lowest.	Medium to High (scalable).	Highest.
Time / Cost to Perform	Fast & Cheap.	Moderate.	Most time-consuming & costly.

Navigating the Cone of Uncertainty



Basis of Estimates

Every estimate must be accompanied by its supporting details: Assumptions, Constraints, Variance Ranges, and Confidence Levels.

The Tale of Two Reserves



Phase 3: Assembling the Architecture (Determine Budget)

Aggregation Engine

Inputs: Activity Cost Estimates, Basis of Estimates, Risk Register.

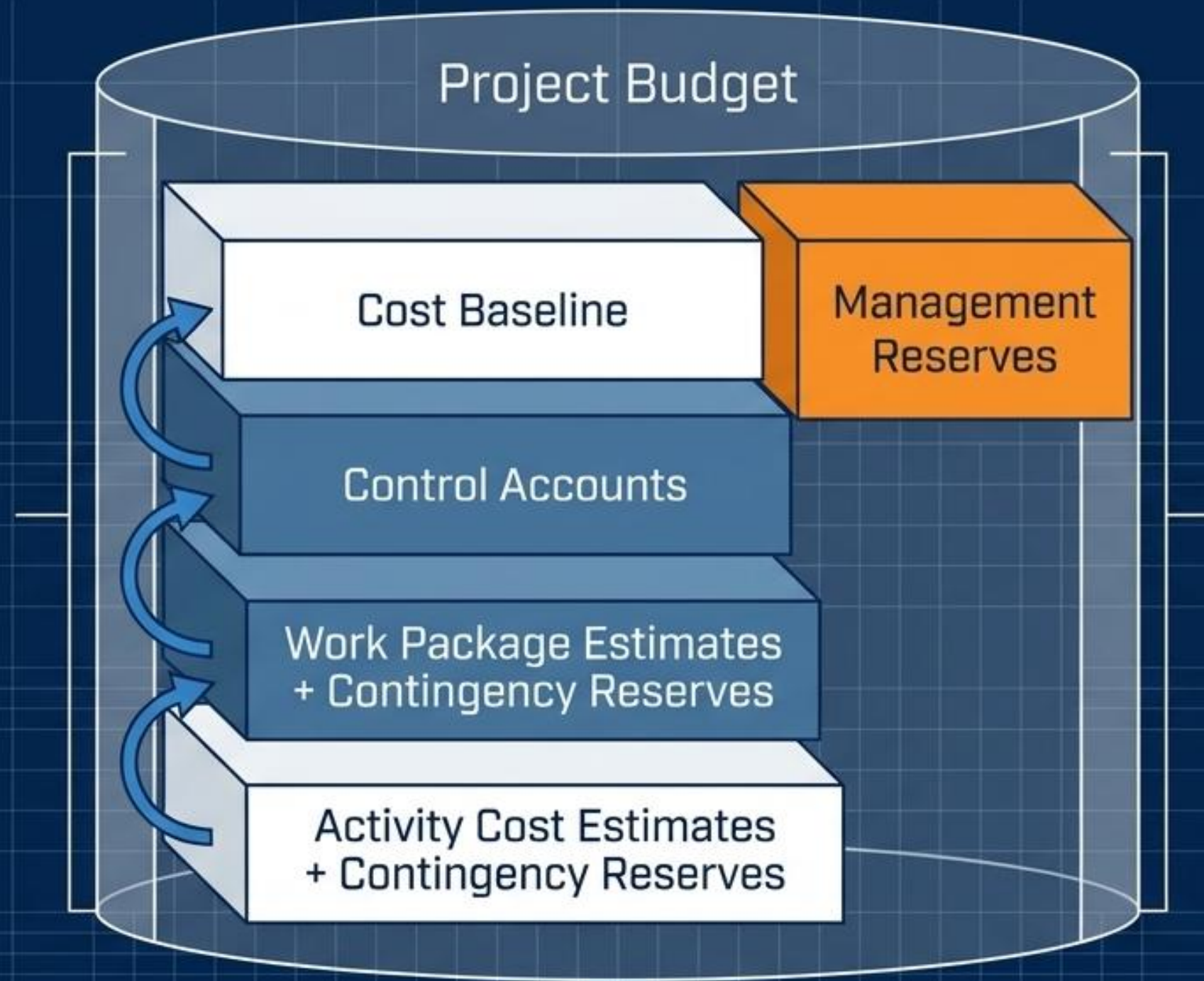
Cost Aggregation Tool: Tallying schedule activity estimates -> Work Packages -> Control Accounts.

EXAM SPOTLIGHT

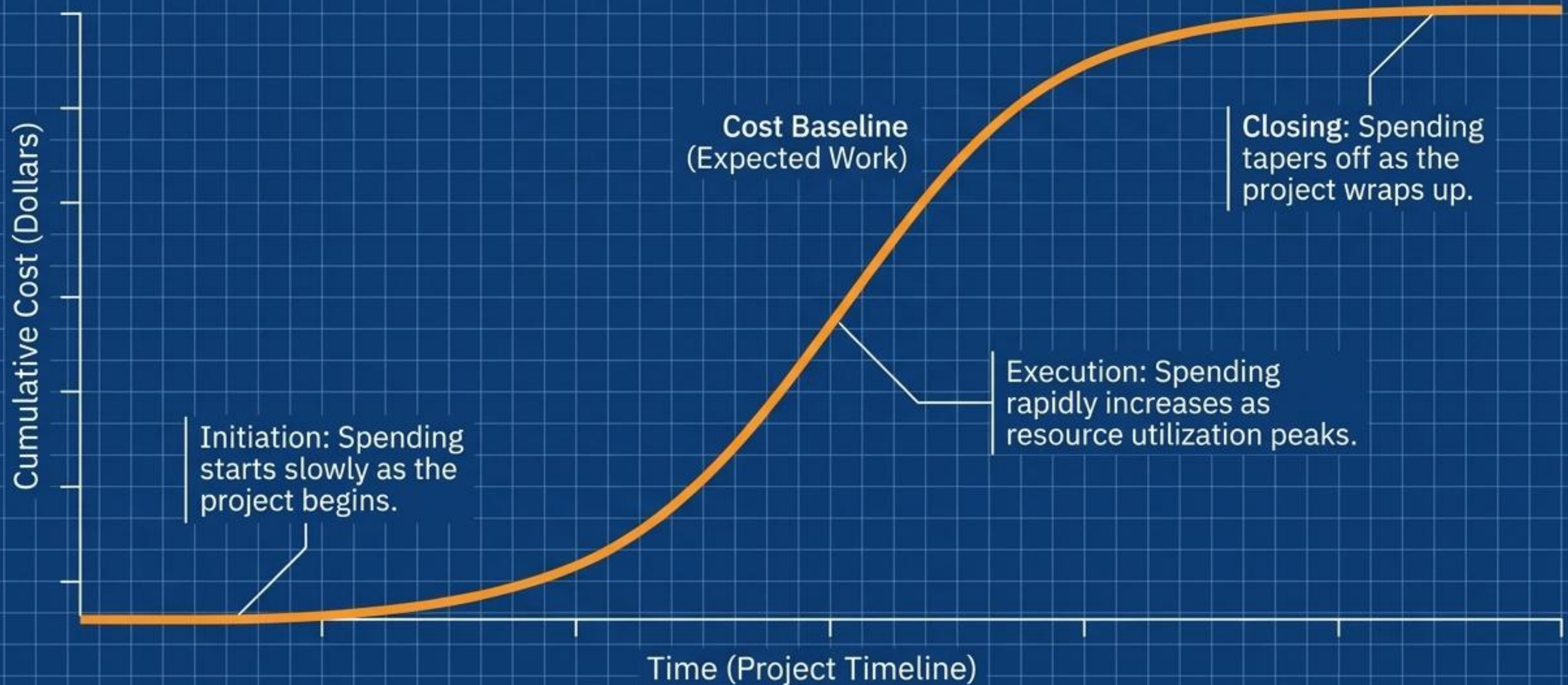
Watch for Indirect Costs! Leases, admin salaries, and shared facilities are often allocated across projects and must be factored into the Basis of Estimates.

Output Goal: Establishing the Authorized Cost Baseline (Performance Measurement Baseline).

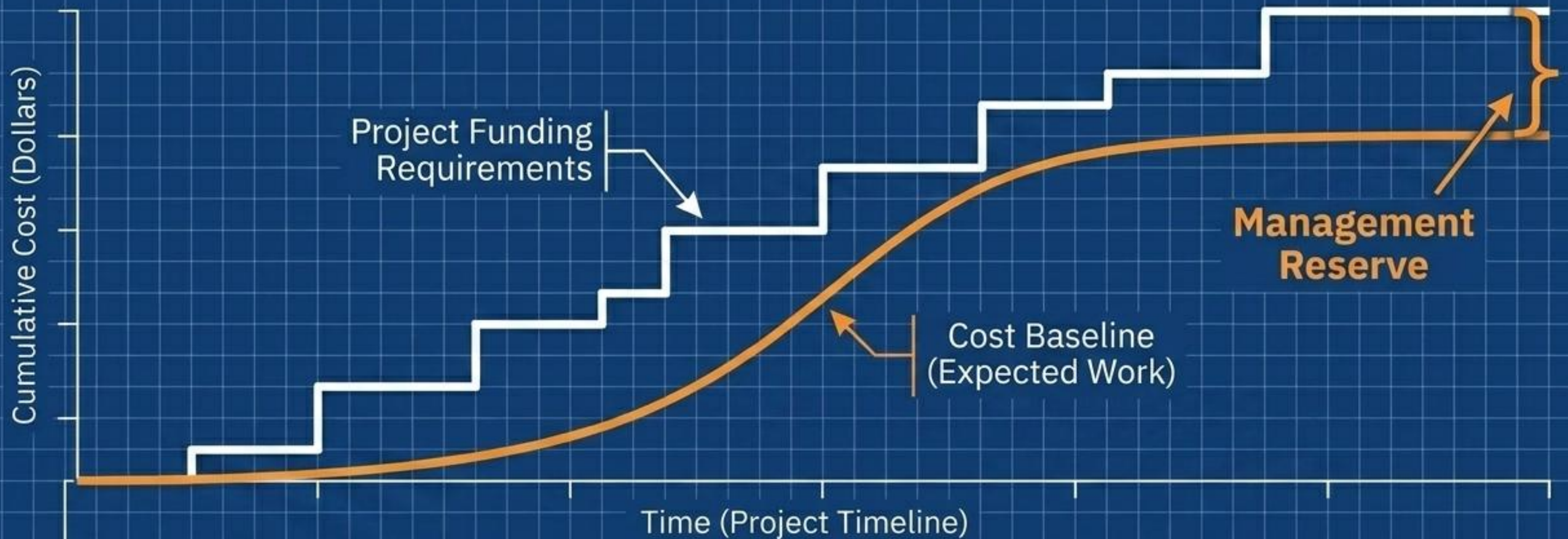
Anatomy of a Project Budget



Mapping the Cost Baseline S-Curve



Funding Limit Reconciliation



Reconciliation ensures the authorized cash flow (stair-steps) always stays ahead of planned expenditures (the S-Curve). If spending outpaces the step, schedule constraints must be imposed to smooth expenses.

Final Blueprint Review: Exam Spotlights

Pricing is NOT Costing.

Estimate Costs is about resource utilization, not profit margins.

Management Reserves are NEVER in the Cost Baseline.

They are part of the overall Project Budget, but cannot be used for Earned Value measurements.

Scope Definition happens early because it is **exponentially easier and cheaper** to influence costs at the beginning of the project.

⚠ Unapproved Baselines are Invalid. The time-phased Cost Baseline must be authorized by the sponsor.